FULL NAME: ______________________________ **BLOCK:** __________

Multiply & Divide Rational Expressions + DOK-3 Closure

Lesson 4-3 · Period 2 · Teacher Edition

OBJECTIVES

Math Objective	Multiply and divide rational expressions by factoring, canceling common factors, and rewriting division as multiplication by the reciprocal; track all domain restrictions through every step.
Language Objective	Justify in writing whether a set is closed under an operation, using sentence frames that cite factoring and cancellation.
Essential Understanding	Rational expressions behave like rational numbers: closed under multiplication and nonzero division, because factoring + cancellation always produces another rational expression.

TOPIC GOALS / ESSENTIAL QUESTION / MATERIALS

Topic Goals	Multiply and divide rational expressions; justify closure under these operations.
Essential Question	Why are rational expressions closed under multiplication and division, and what must we track to prove it?
Materials	Student packet, pencil. Teacher models Example 3; DOK-3 Practice #13 is the capstone closure-argument task.

[PIN] PRE-FLIGHT — P2 IS THE DOK-3 SPINE DAY

Today's conceptual core is Practice #13 (Closure Argument). Students must produce a written paragraph with at least ONE worked example showing a product AND one showing a quotient, concluding that the result in each case is itself rational. That argument IS the lesson's deepest content.
Pace: Try-It 3 (4 min) → #27 (4 min) → Try-It 4 (4 min) → #29 (4 min) → Try-It 5 (3 min) → #32 (4 min) → #13 (12 min). ∼35 min total Explore.
If running tight: cut Try-It 5 or #32. Never cut #13.

Tag: bridge from P1

Do Now **DOK:** 1 **Minutes:** 5

Teacher does

- Hand out at door.
- Circulate 3 min.
- Cold-call 1 student for factoring and domain restriction.

Students do

- Simplify by factoring and canceling.
- State domain restrictions.

Questions to ask

- What do you factor first?
- Is xy constant here? What values are excluded from the domain?

Adult role
Solo teacher: circulate silently. No hints.

Do Now — Simplify this rational expression
What is the simplified form of the rational expression? What is the domain?

$$\frac{x^3 + 9x^2 - 10x}{x^3 - 9x^2 - 10x}$$

[CHECK] ANSWER KEY

Answer key (from source): $\dfrac{(x+10)(x-1)}{(x-10)(x+1)}$, $x \neq -1, 0, 10$.

[TARGET] SENTENCE FRAME

"I factor the numerator as ___ and the denominator as ___. I cancel the common factor ___, leaving ___. The domain excludes $x =$ ___ because that value makes the denominator zero."

Tag: teacher models Example 3

Launch | **DOK:** 2 | **Minutes:** 12

Teacher does

- Project Example 3. Think aloud: factor every numerator and denominator before canceling.
- "For division I flip the second fraction — the RECIPROCAL — before I do anything else."
- Flag the Multiply/Divide Rules card — print it in student minds.
- 30-sec pause: "What domain values did I exclude and why?" (partner check)
- Do NOT solve Try-Its or Practice items — release at minute 17.

Students do

- Watch Example 3 silently.
- During pause: state domain restrictions to partner.
- Copy Rules card into margin.
- Note the division-as-reciprocal step: this comes back in #13.

Questions to ask

- What must I do BEFORE canceling in a division problem?
- Why can't I cancel terms (numerator + denominator) — only factors?
- If $x = 2$ makes ANY denominator zero at ANY step, is it excluded? (Yes, always.)

Adult role
Project. Think aloud. Release promptly at minute 17.

[MIC] TEACHER SCRIPT

1. "Watch me multiply two rational expressions. I factor everything before I cancel."
2. "Step 1: Factor every numerator and denominator completely."
3. "Step 2: For division, rewrite as multiplication by the reciprocal NOW — before factoring or canceling."
4. "Step 3: Cancel only common FACTORS across any numerator with any denominator."
5. "Step 4: State the domain — list every x-value that makes any denominator zero, including the reciprocal's new denominator."
6. "Today's DOK-3 task asks WHY rational expressions are always closed under these operations. You'll write a paragraph with a worked example."
7. Release to Explore.

Tag: TPS + DOK-3 driver

Explore **DOK:** 2-3 **Minutes:** 33

Teacher does

- 5 laps across 33 min.
- Lap 1 (4 min): Try-It 3 — multiply. Press pairs to factor before canceling.
- Lap 2 (4 min): #27 — multiply with factoring. Watch for premature canceling.
- Lap 3 (4 min): Try-It 4 — divide. Check that pairs flip FIRST.
- Lap 4 (4 min): #29 — divide. Confirm domain restriction includes reciprocal's new denominator.
- Lap 5 (12 min): [STAR] #13 Closure Argument. Students write a paragraph with ONE product example AND one quotient example, concluding the result is rational.
- Do NOT give #13 answers. Pairs present argument to each other; one pair presents at Share.

Students do

- 6 items + #13 in order: Try-It 3 → #27 → Try-It 4 → #29 → Try-It 5 → #32 → #13.
- For #13: BEFORE writing, identify what 'closed' means and plan your two examples.
- Justify with sentence frame.

Questions to ask

- Did you factor completely before canceling?
- #27: what common factor appears in numerator and denominator?
- #29: what is the reciprocal of the divisor? What new domain restriction does it add?
- Try-It 5: both multiply and divide steps — which comes first?
- #13: what makes a set 'closed' under an operation? (The result is always in the same set.)
- #13: can you name a product of two rational expressions that is NOT rational? (No — that's the argument.)

Adult role
Solo teacher: 5 laps. Press CLOSURE argument structure on #13.

Try It 3 — Multiply rational expressions

Find the simplified form of each product, and state the domain.

a. $\dfrac{x^2-16}{9-x}\cdot\dfrac{x^2+x-90}{x^2+14x+40}$

b. $\dfrac{x+3}{4x}\cdot\dfrac{3x-18}{6x+18}\cdot\dfrac{x^2}{4x+12}$

[CHECK] ANSWER / ANTICIPATED TALK

Answer key (from source): a. $-(x-4)$ for all x except 9, -10, and -4. b. $\dfrac{x(x-6)}{32(x+3)}$ for all x except 0 and -3.

Practice #27 — Multiply with factoring

Find the product and the domain.

$$\frac{(x-y)^2}{x+y}\cdot\frac{3x+3y}{x^2-y^2}$$

[CHECK] ANSWER / ANTICIPATED TALK

Answer key (from source): $\dfrac{3(x-y)}{x+y}$ for all real numbers except $x=y$ or $x=-y$.

Try It 4 — Divide rational expressions

Find the simplified form of each product and the domain.

a. $\dfrac{x^3-4x}{6x^2-13x-5}\cdot(2x^3-3x^2-5x)$

b. $\dfrac{3x^2+6x}{x^2-49}\cdot(x^2+9x+14)$

[CHECK] ANSWER / ANTICIPATED TALK

Answer key (from source): a. $\dfrac{x^2(x+2)(x-2)(x+1)}{3x+1}$ for all x except $\dfrac{5}{2}$ and $-\dfrac{1}{3}$. b. $\dfrac{3x(x+2)^2}{x-7}$ for all x except 7 and -7.

Practice #29 — Divide: rewrite as multiplication

Find the product and the domain.

$$\frac{2x^2-10x}{(x-5)(x^2-1)}\cdot(3x^2+4x+1)$$

[CHECK] ANSWER / ANTICIPATED TALK

Answer key (from source): $\dfrac{2x(3x+1)}{x-1}$ for all x except -1, 1, and 5.

Try It 5 — Mixed multiply/divide

Find the simplified quotient and the domain of each expression.

a. $\dfrac{1}{x^2+9x} \div \left(\dfrac{6-x}{3x^2-18x}\right)$

b. $\dfrac{2x^2-12x}{x+5} \div \left(\dfrac{x-6}{x+5}\right)$

[CHECK] ANSWER / ANTICIPATED TALK

Answer key (from source): a. $-\dfrac{3}{x+9}$ for all x except 0, -9, and 6. b. $2x$ for all x except 6 and -5.

Practice #32 — Multi-step multiply/divide

Find the quotient and the domain.

$$\frac{25x^2-4}{x^2-9} \div \frac{5x-2}{x+3}$$

[CHECK] ANSWER / ANTICIPATED TALK

Answer key (from source): $\dfrac{5x+2}{x-3}$ for all x except -3, $\dfrac{2}{5}$, and 3.

Practice #13 [STAR] DOK-3 CLOSURE ARGUMENT

Higher Order Thinking Why are rational expressions closed under multiplication and division?

[STAR] DOK-3 CLOSURE ARGUMENT — Practice #13

Students should produce a paragraph with at least ONE worked example showing a product AND one showing a quotient, concluding that the result in each case is itself rational.
Key criteria: (1) cites factoring + cancellation by name, (2) shows domain restriction tracking, (3) explicitly concludes the result is a rational expression.

[CHECK] ANSWER / ANTICIPATED TALK

Answer key (from source): Sample: $\dfrac{a}{b} \cdot \dfrac{c}{d} = \dfrac{ac}{bd}$, and $\dfrac{a}{b} \div \dfrac{c}{d} = \dfrac{a}{b} \cdot \dfrac{d}{c} = \dfrac{ad}{bc}$.

Tag: academic conversation + P3 bridge

Share / Summary **DOK:** 2 **Minutes:** 5

Teacher does

- Cold-call 1 pair to present their Closure Argument using sentence frame.
- Class signals thumbs. Write the closure definition on board.
- P3 bridge: "Next period we apply these skills to more complex items — bring your domain-restriction list habit."

Students do

- Presenting pair reads their paragraph and cites their two worked examples.
- Rest of class signals and writes takeaway in margin.

Questions to ask

- Summarize: what does it mean for rational expressions to be 'closed' under multiplication?
- Does division by zero break closure? How do domain restrictions handle that?

Adult role
Cold-call a pair that struggled on #13 — hold them accountable to the closure-argument structure.

Sentence frames:
Pair share (Closure Argument): "I rewrite the division as multiplication by the reciprocal, giving ___. After factoring, I cancel the factor ___, leaving ___. The domain excludes $x =$ ___ because those values make at least one denominator zero at some step."
Whole class: one pair reads the closure argument. Class signals thumbs up/sideways.

Tag: summary recap (not graded)

Exit Ticket **DOK:** 1-2 **Minutes:** 3

Teacher does

- Collect at door.
- Scan stem 3 (domain restrictions). Plan P3 Do Now if many students miss the reciprocal's denominator.

Students do

- 4 sentence stems, honest.

Questions to ask

- Did students remember to include reciprocal's denominator in domain restrictions?

Adult role
Collect. No grade.

[CLIPBOARD] EXIT TICKET — Student Form

To divide rational expressions I first ___.
I cancel common factors, not ___.
My domain restriction list must include zeros from ___.
One biggest thing I learned today: ______________________________

OPTIONAL REINFORCEMENT (not collected)

If you finish Explore early. #40 is a finance performance task that applies rational expressions to a real-world context.

Optional #1 (Savvas Practice, performance-task finance item)

Performance Task The approximate annual interest rate r of a monthly installment loan is given by the formula:

$$r = \frac{\left[\dfrac{24(nm - p)}{n}\right]}{p + \dfrac{nm}{12}}$$

Part A Find the approximate annual interest rate (to the nearest percent) for a four-year signature loan of \$20,000 that has monthly payments of \$500.
Part B Find the approximate annual interest rate (to the nearest tenth percent) for a five-year auto loan of \$40,000 that has monthly payments of \$750.

[CHECK] ANSWER / ANTICIPATED TALK

Answer key (from source): Part A 9%. Part B 4.6%.

[CLOCK] PERIOD A / PERIOD F — 65-MIN CLASS NOTES

Both sections should have 65 min for P2. Use the extra 10 min on Explore #13 (Closure Argument) — the richer the written paragraph, the better the DOK-3 payoff.
If finished early: hand out Practice #40 (finance performance task) as fast-finisher.

[PUZZLE] MODIFICATIONS / SUPPORT FOR IEP STUDENTS

- Provide the Multiply/Divide Rules card as a sticker/cut-out.
- For #13, provide a paragraph starter: "I will show that the product of two rational expressions is rational. Let $P(x)/Q(x)$ and $R(x)/S(x)$ be two rational expressions…"
- Accept verbal closure argument in lieu of written paragraph.

[GLOBE] SUPPORTS FOR ELL STUDENTS

- Sentence frames on student page (don't skip).
- Word cards: "reciprocal" = flip the fraction; "closed" = the result stays in the same type; "domain restriction" = values x cannot equal.
- "Closure Argument" = a written explanation showing WHY something is always true.
- Pair ELL students with English-dominant partners for TPS.